

Surd rules



1

a False

b True

2

a $\sqrt{252}$

b $\sqrt{224}$

c $\sqrt{45}$

d $\sqrt{1000}$

3

a $4 + 3\sqrt{5}$

b $5\sqrt{2}$

c 45

d 735

e $6\sqrt{x}$

f $\sqrt{3}$

g $\frac{\sqrt{23}}{4}$

Multiplication and division of surds

4

a $\sqrt{22}$

b $4\sqrt{2}$

c $20\sqrt{7}$

d $15\sqrt{35}$

e $\sqrt{70}$

f $35\sqrt{21}$

g $\sqrt{35}$

h $80\sqrt{5}$

i $\sqrt{231}$

j $11\sqrt{5}$

k $20\sqrt{11}$

l $30\sqrt{55}$

m $14\sqrt{11}$

n $24\sqrt{15}$

o $320\sqrt{3}$

p $40\sqrt{51}$

q $340\sqrt{7}$

r $216\sqrt{17}$

5

a $\sqrt{5}$

b $2\sqrt{7}$

c 32

d $8\sqrt{11}$

e $4\sqrt{7}$

f $\sqrt{3}$

g $\sqrt{11}$

h $\sqrt{3}$

i $\sqrt{7}$

j $\sqrt{13}$

k $5\sqrt{7}$

l $10\sqrt{5}$

m $15\sqrt{2}$

n $2\sqrt{7}$

o 3

p 6

q 10

r 16

s 10

t $\frac{5}{9}$

u $\frac{9}{2}$

6

a 2

b $\frac{1}{\sqrt{5}}$

c 4

d $\frac{1}{\sqrt{3}}$

e $\frac{1}{\sqrt{3}}$

f 2

g $\frac{2}{3}$

h $\frac{3}{2}$

i $17\sqrt{7}$



Mixed operations with surds

7

a $26\sqrt{3}$

b $-18\sqrt{5}$

c $27\sqrt{6} + 11\sqrt{13}$

d $20\sqrt{6} + 5\sqrt{11}$

e $4\sqrt{6}$

f $16\sqrt{5}$

g $-3\sqrt{2}$

h $42\sqrt{5}$

8

a $3\sqrt{3}$

b $3\sqrt{3}$

c $5\sqrt{2}$

d $6\sqrt{5}$

9

a $-5\sqrt{7} + \sqrt{35}$

b $7\sqrt{14} + 42\sqrt{2}$

c $-42\sqrt{22} + 7\sqrt{77}$

d $\sqrt{26} - 3\sqrt{13} + 5\sqrt{2} - 15$

e $\sqrt{14} - \sqrt{77} - \sqrt{26} + \sqrt{143}$

f $12 + 2\sqrt{11}$

g $29 + 12\sqrt{5}$

h $\sqrt{77} + 4\sqrt{11}$

i $3\sqrt{7} + \sqrt{21}$

j $-6\sqrt{2} + \sqrt{22}$

k $3\sqrt{39} - 15\sqrt{3}$

l $\sqrt{33} + \sqrt{39}$

m $-4\sqrt{77} + 4\sqrt{14}$

n $15\sqrt{11} + 3\sqrt{55}$

o $-24\sqrt{14} + 8\sqrt{6}$

p $15\sqrt{10} + 20\sqrt{14}$

q $63\sqrt{5}$

r $11\sqrt{15}$

s $24\sqrt{77} - 32\sqrt{55}$

10

a 60

b $\frac{12\sqrt{3}}{\sqrt{5}}$

c $\sqrt{5} + \sqrt{7}$

d $\frac{3\sqrt{3}}{\sqrt{10}}$

11

a $x = 5999, y = -560$

b $x = 5, y = -3$

12

a $42 + (8 + 2m)\sqrt{13} + m^2$

b $m = -4$

Rationalise the denominator

13

a No

b Yes

c No

14

$$\text{a } \frac{\sqrt{11}}{11}$$

$$\text{b } \frac{\sqrt{22} - 2\sqrt{11}}{11}$$



$$\text{c } \frac{7\sqrt{12}}{12}$$

$$\text{d } \frac{14\sqrt{143}}{117}$$

15

$$\text{a } \frac{\sqrt{3}}{3}$$

$$\text{b } 7\sqrt{3}$$

$$\text{c } \frac{-9(\sqrt{5} + 7)}{44}$$

$$\text{d } \frac{4(9\sqrt{11} + 5)}{433}$$

$$\text{e } \frac{9 + 2\sqrt{14}}{5}$$

$$\text{f } \frac{\sqrt{10} - 1}{9}$$

$$\text{g } 4(\sqrt{10} + \sqrt{3})$$

$$\text{h } \frac{5\sqrt{42} - 4\sqrt{14}}{118}$$

16

$$\text{a } -\frac{2\sqrt{2}}{23}$$

$$\text{b } \frac{3\sqrt{5} - 15\sqrt{2}}{20}$$

17 No solution provided

18

$$\text{a } \frac{\sqrt{3} - 1}{2}$$

$$\text{b } 16$$

19

$$\text{a } 19 + 8\sqrt{3}$$

$$\text{b } \frac{6 - \sqrt{3}}{3}$$

$$\text{20 } \sqrt{1800}$$

Applications

$$\text{21 } 12\sqrt{2} \text{ units}$$

$$\text{22 } 3\sqrt{5} \text{ units}$$

23

$$\text{a } 4\sqrt{7} \text{ m}$$

$$\text{b } 3 \text{ m}^2$$

$$\text{24 } c = \sqrt{26} \text{ units}$$

$$\text{25 } 9\sqrt{3} \text{ cm}$$

26

$$\text{a } \frac{\sqrt{5} + 1}{2}$$

$$\text{b } 1.618$$

27 $4\sqrt{10} - 12$ m

